

1 Introduction

The main material of this project is [MR91]. The idea of that paper is to adopt the method in [HK78] and give a new proof of the Alexandrov type theorem for the r -th mean curvature. Moreover, this approach can also be used for hypersurfaces in hyperbolic space and upper semi-sphere, after some modifications. The main work of the report besides typing is filling in some omitted details of [MR91], especially the proof of the spherical case. I sincerely thank Prof. Li for in-depth teaching and those well-arranged notes. It is very lucky for me to take this course.

2 Preliminaries

MEAN CURVATURE

Let $x : M^n \rightarrow R^{n+1}(c)$ be an immersed compact orientable hypersurface, $k_1, \dots, k_n$ the principal curvatures.

Definition 1 (r -th mean curvature). *The r -th mean curvature H_r is defined by*

$$P_n(t) = (1 + tk_1) \cdots (1 + tk_n) = 1 + \binom{n}{1} H_1 t + \cdots + \binom{n}{n} H_n t^n. \quad (1)$$

For example, H_1 is the mean curvature, H_2 , up to a constant, is the scalar curvature, and H_n is the Gauss curvature.

Here is an important lemma from the inequalities in [Går59].

Lemma 2. *Suppose k_i are all positive at some point in M .*

(1) *If $H_r > 0$ everywhere on M , then so is for H_k , $1 \leq k \leq r - 1$.*

(2) *We have*

$$H_k^{\frac{k-1}{k}} \leq H_{k-1}, \quad H_k^{\frac{1}{k}} \leq H_1. \quad (2)$$

Moreover, for $k \geq 2$, the equality holds only at umbilical points.

THE EXISTENCE OF CONVEX POINT

Lemma 3. *Let $x : M^n \rightarrow R^{n+1}(c)$ be an immersed compact orientable hypersurface,*

- (1) For $c = 0$, there is a point in M , where all the principal curvature are positive.
- (2) For $c = 1$, suppose $\text{im } x \subset S_+^{n+1}$, there is a point in M , where all the principal curvature are positive.
- (3) For $c = -1$, there is a point in M , where all the principal curvature are greater than 1.

Remark 4. For example,

- if $c = 0$, since M is compact, we can take a sphere tangent to M ;
- if $c = 1$, take the point where the height $\langle x, a \rangle$ attains maximum;
- if $c = -1$, take the point where distance of H^{n+1} attains maximum.

AN INTEGRATION FORMULA

Let $x : M^n \rightarrow \mathbb{R}^{n+1}(c)$ be an embedded compact hypersurface, N the inner unit normal vector field, Ω the compact domain with $\partial\Omega = M$.

Lemma 5 ([Cha95]). Suppose the volume element of $\mathbb{R}^{n+1}$ has the expression

$$\text{dvol} = \exp_{x(p)}(tN(p)) = F(p, t) \, dt \, dA,$$

then we have an integration formula

$$\int_{\Omega} f \, \text{dvol} = \int_M \int_0^{c(p)} f(\exp_{x(p)}(tN(p))) F(p, t) \, dt \, dA \quad (3)$$

where c is the cut function of M .

3 The Euclidean case

Let $x : M^n \rightarrow \mathbb{R}^{n+1}$ be an immersed compact orientable hypersurface, N the inner unit normal vector field. By direct calculation,

$$\begin{aligned} \Delta \langle x, x \rangle &= 2(D \langle x, e_i \rangle)_i \\ &= 2 \sum_i \delta_{ij} (\langle e_i, e_j \rangle + \langle x, e_\alpha \rangle h_{ij}^\alpha) = 2n(1 + H \langle x, N \rangle). \end{aligned}$$

Using divergence theorem,

$$\int_M (1 + H \langle x, N \rangle) \, dA = 0. \quad (4)$$

Lemma 6 (Minkowski formulae). *Let $x : M^n \rightarrow \mathbb{R}^{n+1}$ be an immersed compact orientable hypersurface, N the inner unit normal vector field, then for $1 \leq r \leq n$, we have*

$$\int_M (H_{r-1} + H_r \langle x, N \rangle) dA = 0. \quad (5)$$

Proof. (From [Hsi56]) For small number t , consider hypersurface

$$x_t(p) = \exp_{x(p)}(-tN(p)) = x(p) - tN(p). \quad (6)$$

Since t is small, N is also a unit normal vector field for x_t , and the principal directions are given by

$$x_{t,*}e_i = (1 + tk_i)e_i, 1 \leq i \leq n. \quad (7)$$

where e_i are the principal directions for x . Thus $(1 + tk_i)k_i(t) = k_i$. For the area element,

$$dA_t = (1 + tk_1) \cdots (1 + tk_n) dA = P_n(t) dA. \quad (8)$$

For the mean curvature,

$$H_1(t) = \frac{1}{n} \sum \frac{k_i}{1 + tk_i} = \frac{P'_n(t)}{nP_n(t)}. \quad (9)$$

So we have by (4), (8), (9)

$$\begin{aligned} 0 &= \int_M n(1 + H_1(t) \langle x, N \rangle) dA_t \\ &= \int_M (nP_n(t) + P'_n(t) \langle x - tN, N \rangle) dA \\ &= \sum_i \int_M n \binom{n}{i} H_i t^i + \binom{n}{i} i H_i t^{i-1} (\langle x, N \rangle - t) dA. \end{aligned} \quad (10)$$

Regarding both sides as polynomials of t , we can solve

$$\int_M (H_{r-1} + H_r \langle x, N \rangle) dA = 0 \quad (11)$$

for $1 \leq r \leq n$. □

Theorem 7 (Heintze-Karcher inequality [HK78, MR91]). *Let $x : M^n \rightarrow \mathbb{R}^{n+1}$ be an embedded compact hypersurface. If $H_1 > 0$ everywhere on M , then we have*

$$\int_M \frac{1}{H_1} dA \geq (n+1) \text{vol}(\Omega) \quad (12)$$

where Ω is the compact domain with $\partial\Omega = M$. Moreover, the equality holds if and only if M^n is a round sphere.

Proof. Recall that $x_t = \exp_{x(p)}(tN(p)) = x(p) + tN(p)$ here, we have

$$dV(x + tN) = (1 - tk_1) \cdots (1 - tk_n) dt dA. \quad (13)$$

Using (3) for $f(x) \equiv 1$,

$$\text{vol}(\Omega) = \int_M \int_0^{c(p)} (1 - tk_1) \cdots (1 - tk_n) dt dA. \quad (14)$$

Note that $c(p) \leq \frac{1}{k_{\max}} \leq \frac{1}{H_1(p)}$ since the normal geodesic is well-defined before reaching the focal point. And as an algebraic inequality,

$$(1 - tk_1) \cdots (1 - tk_n) \leq (1 - tH_1)^n \quad (15)$$

Then from (14),

$$\text{vol}(\Omega) \leq \int_M \int_0^{\frac{1}{H_1}} (1 - tH_1)^n dt dA = \frac{1}{n+1} \int_M \frac{1}{H_1} dA. \quad (16)$$

The equality holds if (15) holds, which means M is totally umbilical. $\square$

Theorem 8 ([MR91]). *Let $x : M^n \rightarrow \mathbb{R}^{n+1}$ be an embedded compact hypersurface. If H_r is constant for some $1 \leq r \leq n$, then M is a round sphere.*

Proof. From Lemma 3, there is a convex point in M , thus H_r is a positive constant. Let Ω be the compact domain with $\partial\Omega = M$. Using Lemma 2, we have $H_r^{\frac{1}{r}} \leq H_1, H_{r-1} \geq H_r^{\frac{r-1}{r}}$. Together with Lemma 6, we get

$$\begin{aligned} 0 &= \int_M (H_{r-1} + H_r \langle x, N \rangle) dA \\ &\geq \int_M (H_r^{\frac{r-1}{r}} + H_r \langle x, N \rangle) dA \\ &= H_r^{\frac{r-1}{r}} \int_M (1 + H_r^{\frac{1}{r}} \langle x, N \rangle) dA. \end{aligned} \quad (17)$$

Recall that from divergence theorem,

$$(n+1)\text{vol}(\Omega) + \int_M \langle x, N \rangle dA = 0. \quad (18)$$

So

$$0 \geq \text{area}(M) - (n+1)H_r^{\frac{1}{r}}\text{vol}(\Omega). \quad (19)$$

Using Theorem 7,

$$(n+1)H_r^{\frac{1}{r}}\text{vol}(\Omega) \leq \text{area}(M). \quad (20)$$

Thus the equality in (20) holds, and hence M is totally umbilical by the rigidity of Theorem 7. $\square$

4 The hyperbolic case

Let $\mathbb{R}_1^{n+2}$ be the real vector space $\mathbb{R}^{n+2}$ endowed with the Lozrentzain metric

$$\langle x, y \rangle = -x_0 y_0 + x_1 y_1 + \cdots + x_{n+1} y_{n+1}.$$

The hyperbolic space $R^{n+1}(-1)$ can be regarded as $H^{n+1} = \{x \in R_1^{n+2} \mid |x|^2 = -1, x_0 \geq 1\}$, with the induced positive-definite metric. Then an immersed compact orientable hypersurface $x : M \rightarrow H^{n+1}$ can be viewed as $x : M \rightarrow R_1^{n+2}$ with $|x|^2 = -1, x_0 \geq 1$. Let N be the inner unit normal vector field, $a \in R_1^{n+2}$. By direct calculation,

$$\begin{aligned} \Delta \langle x, a \rangle &= (D \langle e_i, a \rangle)_i \\ &= \sum_i \delta_{ij} (\langle e_i, a \rangle + \langle e_\alpha, a \rangle h_{ij}^\alpha) = n(\langle x, a \rangle + H_1 \langle N, a \rangle). \end{aligned}$$

Using divergence theorem,

$$\int_M (\langle x, a \rangle + H_1 \langle N, a \rangle) dA = 0. \quad (21)$$

Lemma 9 ([MR91]). *Let $x : M^n \rightarrow H^{n+1}$ be an immersed compact orientable hypersurface, N the inner unit normal vector field, then for $1 \leq r \leq n$ and arbitrary $a \in \mathbb{R}_1^{n+2}$, we have*

$$\int_M (H_{r-1} \langle x, a \rangle + H_r \langle N, a \rangle) dA = 0. \quad (22)$$

Proof. For small number t , consider hypersurface

$$x_t(p) = \exp_{x(p)}(-tN(p)) = x(p) \cosh t - N(p) \sinh t. \quad (23)$$

By solving Jacobi field equation, the unit normal vector field is given by $N_t = -x \sinh t + N \cosh t$, and the principal directions are given by

$$x_{t,*} e_i = (\cosh t + k_i \sinh t) e_i, 1 \leq i \leq n \quad (24)$$

where e_i are the principal directions for x . Thus $(\cosh t + k_i \sinh t) k_i(t) = \sinh t + k_i \cosh t$. For the area element,

$$\begin{aligned} dA_t &= (\cosh t + k_1 \sinh t) \cdots (\cosh t + k_n \sinh t) dA \\ &= \cosh^n t \cdot P_n(\tanh t) dA. \end{aligned} \quad (25)$$

For the mean curvature,

$$\begin{aligned} H_1(t) &= \frac{1}{n} \sum \frac{\sinh t + k_i \cosh t}{\cosh t + k_i \sinh t} = \frac{1}{n} \sum \frac{\tanh t + k_i}{1 + k_i \tanh t} \\ &= \frac{n \cosh t \sinh t \cdot P_n(\tanh t) + P'_n(\tanh t)}{n \cosh^2 t \cdot P_n(\tanh t)}. \end{aligned} \quad (26)$$

So we have by (21), (25), (26),

$$\begin{aligned}
0 &= \int_M n(\langle x_t, a \rangle + H_1(t)\langle N_t, a \rangle) dA_t \\
&= \int_M n \cosh^2 t \cdot P_n(\tanh t) \langle x \cosh t - N \sinh t, a \rangle dA \\
&\quad + \int_M (n \cosh t \sinh t \cdot P_n(\tanh t) + P'_n(\tanh t)) \langle -x \sinh t + N \cosh t, a \rangle dA \\
&= \int_M (nP_n(\tanh t) \langle x, a \rangle + P'_n(\tanh t) \langle -x \tanh t + N, a \rangle) dA.
\end{aligned} \tag{27}$$

Regarding both sides as polynomials of $\tanh t$, we can solve

$$\int_M (H_{r-1} \langle x, a \rangle + H_r \langle N, a \rangle) dA = 0 \tag{28}$$

for $1 \leq r \leq n$. $\square$

Definition 10. We define a positive function $\rho_n : (1, \infty) \rightarrow (0, \infty)$ with parameter $n \in \mathbb{N}$ by

$$\rho_n(u) = \int_0^{\coth^{-1} u} (\cosh t - u \sinh t)^n \cosh t dt. \tag{29}$$

We have the following Heintze-Karcher type inequality.

Theorem 11 ([MR91]). Let $x : M^n \rightarrow H^{n+1}$ be an embedded compact hypersurface. If $H_r > 1$ everywhere on M , then we have

$$\int_M (\langle x, a \rangle + H_r^{\frac{1}{r}} \langle N, a \rangle) \rho_n(H_r^{\frac{1}{r}}) dA \geq 0 \tag{30}$$

for $a \in \mathbb{R}_1^{n+2}$ with $|a|^2 = -1$. Moreover, the equality holds if and only if M is a geodesic sphere.

Proof. Recall that $x_t = \exp_{x(p)}(tN(p)) = x(p) \cosh t + N(p) \sinh t$ here, $dV(x \cosh t + N \sinh t) = (\cosh t - k_1 \sinh t) \cdots (\cosh t - k_n \sinh t) dt dA$. $\tag{31}$

Note $\overline{\Delta} \langle x, a \rangle = (n+1) \langle x, a \rangle$, and $\overline{\nabla} \langle x, a \rangle = a$, from divergence theorem,

$$(n+1) \int_{\Omega} \langle x, a \rangle dV + \int_M \langle N, a \rangle dA = 0. \tag{32}$$

Using (3) for $f(x) = (n+1) \langle x, a \rangle$,

$$\begin{aligned}
- \int_M \langle N, a \rangle dA &= (n+1) \int_M \int_0^{c(p)} \langle x_t, a \rangle \prod_i (\cosh t - k_i \sinh t) dt dA.
\end{aligned} \tag{33}$$

From Lemma 3, there is a point in M , where all the principal curvatures are greater than 1. Using Lemma 2, we have $1 < H_r^{\frac{1}{r}} \leq H_1$. Note that $c(p) \leq \coth^{-1} k_{\max} \leq \coth^{-1} H_1(p) \leq \coth^{-1} H_r^{\frac{1}{r}}(p)$. And as an algebraic inequality,

$$\prod_i (\cosh t - k_i \sinh t) \leq (\cosh t - H_1 \sinh t)^n \leq (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n. \quad (34)$$

Then from (33),

$$\begin{aligned} & -\frac{1}{n+1} \int_M \langle N, a \rangle dA \\ & \leq \int_M \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n \langle x_t, a \rangle dt dA. \end{aligned} \quad (35)$$

On the other hand, by taking $w = \cosh t - H_r^{\frac{1}{r}} \sinh t$, we can show

$$\begin{aligned} & (n+1) \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n (\sinh t - H_r^{\frac{1}{r}} \cosh t) dt \\ & = \int_1^0 dw^{n+1} = -1. \end{aligned} \quad (36)$$

So multiplying by $\langle N, a \rangle$ and integrating over M , we have

$$\begin{aligned} & -\frac{1}{n+1} \int_M \langle N, a \rangle dA \\ & = \int_M \langle N, a \rangle \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n (\sinh t - H_r^{\frac{1}{r}} \cosh t) dt dA. \end{aligned} \quad (37)$$

Putting together (35) and (37),

$$\begin{aligned}
0 &\leq \int_M \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n \langle x_t, a \rangle dt dA \\
&\quad - \int_M \langle N, a \rangle \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n (\sinh t - H_r^{\frac{1}{r}} \cosh t) dt dA \\
&= \int_M \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n \\
&\quad \cdot (\langle x \cosh t + N \sinh t, a \rangle - (\sinh t - H_r^{\frac{1}{r}} \cosh t) \langle N, a \rangle) dt dA \\
&= \int_M (\langle x, a \rangle + H_r^{\frac{1}{r}} \langle N, a \rangle) \int_0^{\coth^{-1} H_r^{\frac{1}{r}}} (\cosh t - H_r^{\frac{1}{r}} \sinh t)^n dt dA \\
&= \int_M (\langle x, a \rangle + H_r^{\frac{1}{r}} \langle N, a \rangle) \rho_n(H_r^{\frac{1}{r}}) dA.
\end{aligned} \tag{38}$$

The equality holds if (34) holds, which means M is totally umbilical. $\square$

Theorem 12 ([MR91]). *Let $x : M^n \rightarrow H^{n+1}$ be an embedded compact hypersurface. If H_r is constant for some $1 \leq r \leq n$, then M is a geodesic hypersphere.*

Proof. From Lemma 3, there is a point in M , where all the principal curvatures are greater than 1, thus H_r is a constant greater than 1. Then $\rho_n(H_r^{\frac{1}{r}})$ is a positive constant. Using Lemma 2, $H_{r-1} \geq H_r^{\frac{r-1}{r}}$. Together with Lemma 9, we get

$$\begin{aligned}
0 &= \int_M (H_{r-1} \langle x, a \rangle + H_r \langle N, a \rangle) dA \\
&\geq \int_M (H_r^{\frac{r-1}{r}} \langle x, a \rangle + H_r \langle N, a \rangle) dA \\
&= H_r^{\frac{r-1}{r}} \int_M (\langle x, a \rangle + H_r^{\frac{1}{r}} \langle N, a \rangle) dA.
\end{aligned} \tag{39}$$

Using Theorem 11,

$$0 \leq \int_M (\langle x, a \rangle + H_r^{\frac{1}{r}} \langle N, a \rangle) dA. \tag{40}$$

Thus the equality in (40) holds, and hence M is totally umbilical by the rigidity of Theorem 11. $\square$

5 The spherical case

Let $x : M^n \rightarrow S^{n+1}$ be an immersed compact orientable hypersurface, N the inner unit normal vector field, $a \in \mathbb{R}^{n+2}$. By direct calculation,

$$\begin{aligned}\Delta \langle x, a \rangle &= (D \langle e_i, a \rangle)_i \\ &= \sum_i \delta_{ij} (\langle e_i, a \rangle + \langle e_\alpha, a \rangle h_{ij}^\alpha) = n(\langle x, a \rangle - H_1 \langle N, a \rangle).\end{aligned}$$

Using divergence theorem,

$$\int_M (\langle x, a \rangle - H_1 \langle N, a \rangle) dA = 0. \quad (41)$$

Lemma 13 ([MR91, Biv83]). *Let $x : M^n \rightarrow S^{n+1}$ be an immersed compact orientable hypersurface, N the inner unit normal vector field, then for $1 \leq r \leq n$ and arbitrary $a \in \mathbb{R}^{n+2}$, we have*

$$\int_M (H_{r-1} \langle x, a \rangle - H_r \langle N, a \rangle) dA = 0. \quad (42)$$

Proof. For small number t , consider hypersurface

$$x_t(p) = \exp_{x(p)}(-tN(p)) = x(p) \cos t - N(p) \sin t. \quad (43)$$

By solving Jacobi field equation, the unit normal vector field is given by $N_t = x \sin t + N \cos t$, and the principal directions are given by

$$x_{t,*} e_i = (\cos t - k_i \sin t) e_i, \quad 1 \leq i \leq n \quad (44)$$

where e_i are the principal directions of x . Thus $(\cos t - k_i \sin t) k_i(t) = \sin t - k_i \cos t$. For the area element,

$$\begin{aligned}dA_t &= (\cos t - k_1 \sin t) \cdots (\cos t - k_n \sin t) dA \\ &= \cos^n t P_n(-\tan t) dA.\end{aligned} \quad (45)$$

For the mean curvature,

$$\begin{aligned}H_1(t) &= \frac{1}{n} \sum \frac{\sin t - k_i \cos t}{\cos t - k_i \sin t} = \frac{1}{n} \sum \frac{\tan t - k_i}{1 - k_i \tan t} \\ &= - \frac{n \cos t \sin t \cdot P_n(-\tan t) + P'_n(-\tan t)}{n \cos^2 t \cdot P_n(-\tan t)}.\end{aligned} \quad (46)$$

So we have by (41), (45), (46),

$$\begin{aligned}
0 &= \int_M n(\langle x_t, a \rangle + H_1 \langle N_t, a \rangle) dA_t \\
&= \int_M n \cos^2 t \cdot P_n(-\tan t) \langle x \cos t - N \sin t, a \rangle dA \\
&\quad + \int_M (-n \cos t \sin t \cdot P_n(-\tan t) - P'_n(-\tan t)) \langle x \sin t + N \cos t \rangle dA \\
&= \int_M (nP_n(-\tan t) \langle x, a \rangle - P'_n(-\tan t) \langle x \tan t + N, a \rangle) dA.
\end{aligned} \tag{47}$$

Regarding both sides as polynomials of $-\tan t$, we can solve

$$\int_M (H_{r-1} \langle x, a \rangle - H_r \langle N, a \rangle) dA = 0 \tag{48}$$

for $1 \leq r \leq n$. $\square$

Definition 14. We define a positive function $\tau_n : (0, \infty) \rightarrow (0, \infty)$ with parameter $n \in \mathbb{N}$ by

$$\tau_n(u) = \int_0^{\cot^{-1} u} (\cos t - u \sin t)^n \cos t dt. \tag{49}$$

We have the following Heintze-Karcher type inequality.

Theorem 15 ([MR91]). Let $x : M^n \rightarrow S_+^{n+1}$ be an embedded compact hypersurface lying in the upper semi-sphere. If $H_r > 0$ everywhere on M , then we have

$$\int_M (\langle x, a \rangle - H_r^{\frac{1}{r}} \langle N, a \rangle) \tau_n(H_r^{\frac{1}{r}}) dA \geq 0 \tag{50}$$

where a is the north pole of S^{n+1} . Moreover, the equality holds if and only if M is umbilical.

Proof. Recall that $x_t = \exp_{x(p)}(tN(p)) = x(p) \cos t + N(p) \sin t$ here,

$$dV(x \cos t + N \sin t) = (\cos t - k_1 \sin t) \cdots (\cos t - k_n \sin t) dt dA. \tag{51}$$

Note $\overline{\Delta} \langle x, a \rangle = -(n+1) \langle x, a \rangle$, and $\overline{\nabla} \langle x, a \rangle = a$, from divergence theorem,

$$(n+1) \int_\Omega \langle x, a \rangle dV = \int_M \langle N, a \rangle dA. \tag{52}$$

Using (3) for $f(x) = (n+1) \langle x, a \rangle$,

$$\int_M \langle N, a \rangle dA = (n+1) \int_M \int_0^{c(p)} \langle x_t, a \rangle \prod_i (\cos t - k_i \sin t) dt dA. \tag{53}$$

From Lemma 3, there is a convex point in M . Using Lemma 2, we have $0 < H_r^{\frac{1}{r}} \leq H_1$. Note that $c(p) \leq \cot^{-1} k_{\max} \leq \cot^{-1} H_1(p) \leq \cot^{-1} H_r^{\frac{1}{r}}(p)$. And as an algebraic inequality

$$\prod_i (\cos t - k_i \sin t) \leq (\cos t - H_1 \sin t)^n \leq (\cos t - H_r^{\frac{1}{r}} \sin t)^n. \quad (54)$$

Then from (53),

$$\frac{1}{n+1} \int_M \langle N, a \rangle dA \leq \int_M \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n \langle x_t, a \rangle dt dA. \quad (55)$$

On the other hand, by taking $w = \cos t - H_r^{\frac{1}{r}} \sin t$, we can show

$$\begin{aligned} & (n+1) \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n (-\sin t - H_r^{\frac{1}{r}} \cos t) dt \\ &= \int_1^0 dw^{n+1} = -1. \end{aligned} \quad (56)$$

So multiplying by $\langle N, a \rangle$ and integrating over M , we have

$$\begin{aligned} & -\frac{1}{n+1} \int_M \langle N, a \rangle dA \\ &= \int_M \langle N, a \rangle \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n (-\sin t - H_r^{\frac{1}{r}} \cos t) dt dA \end{aligned} \quad (57)$$

Putting together (55) and (57),

$$\begin{aligned} 0 &\leq \int_M \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n \langle x_t, a \rangle dt dA \\ &+ \int_M \langle N, a \rangle \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n (-\sin t - H_r^{\frac{1}{r}} \cos t) dt dA \\ &= \int_M \int_0^{\cot^{-1} H_r^{\frac{1}{r}}} (\cos t - H_r^{\frac{1}{r}} \sin t)^n \\ &\quad \cdot (\langle x \cos t + N \sin t, a \rangle + (-\sin t - H_r^{\frac{1}{r}} \cos t) \langle N, a \rangle) dt dA \\ &= \int_M (\langle x, a \rangle - H_r^{\frac{1}{r}} \langle N, a \rangle) \tau_n(H_r^{\frac{1}{r}}) dA. \end{aligned} \quad (58)$$

The equality holds if (54) holds, which means M is totally umbilical. $\square$

Theorem 16 ([MR91]). *Let $x : M^n \rightarrow S_+^{n+1}$ be an embedded compact hypersurface. If H_r is constant for some $1 \leq r \leq n$, then M is a geodesic hypersphere.*

Proof. From Lemma 3, there is a convex point in M , thus H_r is a positive constant. Then $\tau_n(H_r^{\frac{1}{r}})$ is a positive constant. Using Lemma 2, $H_{r-1} \geq H_r^{\frac{r-1}{r}}$. Together with Lemma 13, we get

$$\begin{aligned} 0 &= \int_M (H_{r-1} \langle x, a \rangle - H_r \langle N, a \rangle) \, dA \\ &\geq \int_M (H_r^{\frac{r-1}{r}} \langle x, a \rangle - H_r \langle N, a \rangle) \, dA \\ &= H_r^{\frac{r-1}{r}} \int_M (\langle x, a \rangle - H_r^{\frac{1}{r}} \langle N, a \rangle) \, dA. \end{aligned} \tag{59}$$

Using Theorem 15,

$$0 \leq \int_M (\langle x, a \rangle - H_r^{\frac{1}{r}} \langle N, a \rangle) \, dA. \tag{60}$$

Thus the equality (60) holds, and hence M is totally umbilical by the rigidity of Theorem 15. $\square$

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